| PS No. | 31 |
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| **Project Title** | **Population Data Visualizer** |
| **Problem Statement** | Analyse city population data from a CSV file to find the total population, the largest city and each city's growth between years, and visualise the results with charts.<br>**Language:** Python 3 **Concepts:** Pandas, Matplotlib, Data Analysis<br>**Note:** CSV columns: City, Year, Population. Growth % = (latest − earliest) ÷ earliest × 100. Use dummy data only. |
| **Test Cases / Case Statements (4 cases)** | |

| INPUT | EXPECTED OUTPUT |
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| **Test Case 1: Normal Input** | |
| population.csv:<br>Raipur,2020,1000000<br>Raipur,2024,1100000<br>Durg,2020,500000<br>Durg,2024,600000<br>Menu option 1 (Summary) | Total population (2024): 1700000<br>Largest city: Raipur (1100000)<br>Fastest growing: Durg (+20.00%) |
| **Test Case 2: Boundary / Minimum Input** | |
| population.csv: 1 row only<br>Raipur,2024,1100000<br>Menu option 1 (Summary) | Total population: 1100000 \| Largest city: Raipur<br>Growth: not available (only one year of data) |
| **Test Case 3: Invalid Input** | |
| population.csv:<br>Raipur,2020,1000000<br>Durg,2020,abc<br>Bhilai,2020,-500 | Row 2 skipped: invalid Population 'abc'.<br>Row 3 skipped: Population must be positive.<br>Valid rows: 1 |
| **Test Case 4: Different Valid Scenario** | |
| Same CSV as Test Case 1<br>Menu option 2 (Growth and charts) | Raipur: +10.00% \| Durg: +20.00%<br>Bar chart 'Population Growth by City' and line chart 'Population Trend' displayed |

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| **Bare Minimum Requirements** | 1. Write the solution in Python 3 (one .py file) and run it from the console.<br>2. Load the CSV file using Pandas and clean invalid or duplicate rows.<br>3. Use groupby() to find the latest population and the growth percentage of every city.<br>4. Show the total population, the largest city and the fastest growing city.<br>5. Plot at least two Matplotlib charts (line and bar) with title and labels. |
| **Instruction** | 1. Understand the problem statement and define the problem clearly.<br>2. Develop the solution using proper logic and programming concepts.<br>3. Implement all required features and pass the given test cases.<br>4. Test and debug the project to ensure correct output.<br>5. Prepare the documentation/report and upload the code and report to GitHub.<br>6. Complete and submit the project within the given deadline. |